

DIALOGUE-BASED STREAMING AUDIO-VISUAL TARGET SPEAKER EXTRACTION WITH PREDICTIVE DIALOGUE INFORMATION

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ABSTRACT

In face-to-face, real-time communication, a talking agent must track the target speaker through natural pauses, turn-taking, and backchannels, often amid background cross-talk. Most target speaker extraction (TSE) studies, however, rely on simulated mixtures with full or sparse overlap and ignore the turn-taking of real conversations. We therefore introduce, to our knowledge, the first benchmark for online audio-visual TSE (AV-TSE), built from intact dyadic interactions with independent third-party interference. Observing that anticipating upcoming activity from semantic, acoustic, and facial cues benefits online AV-TSE, we propose an LLM-based target-speaker voice activity projection (TS-VAP) module. Unlike conventional VAP with separated speaker channels, it forecasts the future activity of the target and conversational partner directly from the overlapping mixture, drawing on the linguistic and conversational knowledge of a speech-LLM, and uses this prediction to guide a low-latency separator. We further combine this predictive context with historical and synchronous speaker context. Experiments show that TS-VAP consistently improves streaming extraction across multiple AV-TSE backbones, and that further combining historical, synchronous, and predictive context yields nearly 1 dB gain on real AV conversations. Project page: <https://jjjjiaozi.github.io/TS-VAP/>.

Index Terms— Audio-visual target speaker extraction, Speech LLM, Voice activity projection, Streaming inference

1. INTRODUCTION

Real-world conversations often unfold in multi-speaker environments, where cross-talk is captured alongside the target speech. Just like humans who listen to one voice at a time, conversational agents are expected to have such attentive ability.

We consider a natural dyadic conversation with an additional unrelated third-party speaker, as illustrated in Fig. 1. Real conversations of this kind are increasingly face-to-face and interactive, as in GPT-4o-style real-time voice assistants [1], where a camera naturally observes the participant. In such settings the face track is the most suitable target cue: it fixes the participant’s identity without a separate enrollment utterance, and because it is visual, it remains reliable even when the acoustic channel is noisy or heavily overlapped. Since the target must also be recovered as the conversation unfolds, this naturally leads to online AV-TSE [2, 3, 4, 5, 6, 7].

The central difficulty lies not in the cue but in the conversation itself. Most online AV-TSE studies evaluate on fully overlapped or manually simulated sparse-overlap mixtures [6, 8, 9], which combine utterances independently and lack the turn-taking structure of

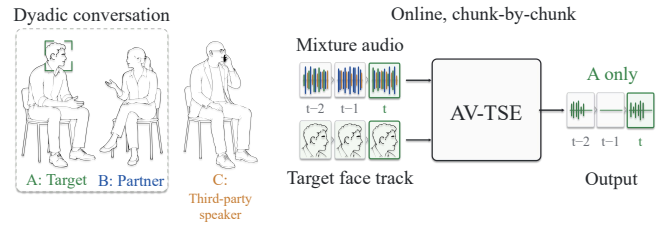


Fig. 1. Online AV-TSE under third-party interference. A and B form a natural dyadic conversation, while C is an unrelated speaker. Given the mixture audio and A’s face track, the system extracts only A’s speech, suppressing both B and C.

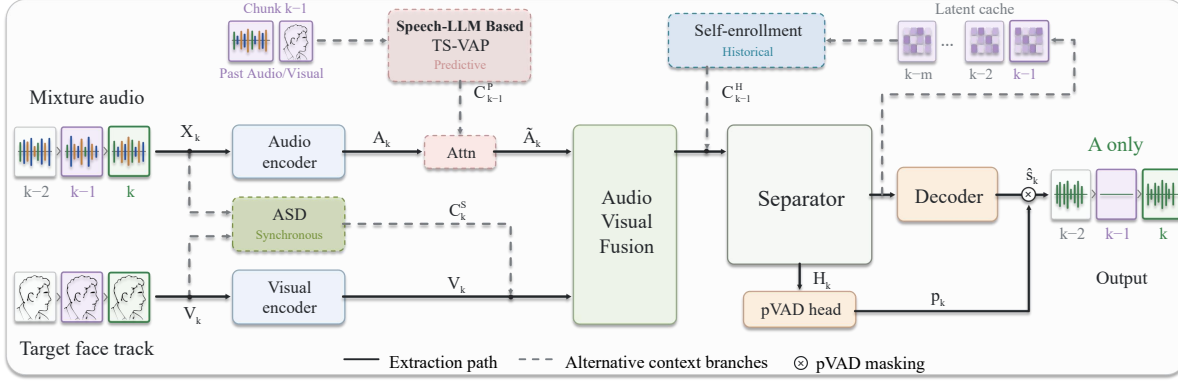
real conversations—structure that governs when the target falls silent and where extraction is hardest. This is especially limiting online: a streaming model sees no future frames and is thus most ambiguous at turn boundaries and under similar-sounding, overlapping voices. Predictive knowledge of upcoming activity would relieve this ambiguity. Voice activity projection (VAP) can forecast future speaker activity from audio [10, 11, 12, 13], but is trained only on clean, single-speaker channels—not the cocktail setting, where turns must be anticipated from one overlapping mixture, which is exactly the regime of online AV-TSE.

We supply this predictive knowledge through an LLM-based TS-VAP module. Beyond the clean-channel limitation above, conventional VAP models anticipate turns mainly from acoustic evidence such as pauses, energy, and prosody. However, linguistic and conversational knowledge that governs turn-taking at the level of meaning remains underexplored: a question projects an answer, a completed clause invites a hand-off, and a backchannel does not. Large language models (LLMs) acquire this knowledge through pretraining. Previous studies have shown that text LLMs supply transcript-level linguistic knowledge for speech extraction [14, 15], but have not explicitly modeled dialogue-level turn-taking knowledge. We therefore adapt the speech-LLM Mini-Omni2 [16] into a TS-VAP that forecasts near-future target and interferer activity directly from the overlapping mixture, anticipating who will speak next from what is being said rather than from acoustic rhythm alone. We use this forecast to guide a low-latency separator, and further situate it alongside two other forms of dialogue context: active speaker detection (ASD) features as synchronous context [8], and self-enrolled speech as historical context [17].

Our contributions are threefold. (1) We introduce, to our knowledge, the first benchmark for online AV-TSE on real dyadic dialogues under third-party interference, built from IEMOCAP [18] and a RealTalk-based test set [19], whose mixtures retain natural turn-taking rather than independently assembled utterances. (2) We pro-

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(a) AV-TSE with target-conditioned activity context



(b) Dialogue cues for extracting target A in chunk k

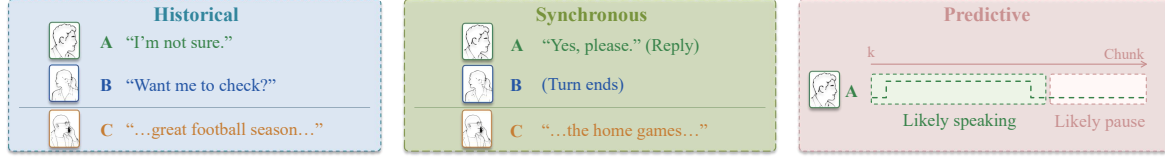


Fig. 2. (a) AV-TSE extracts target speech in chunk k . Dashed branches denote alternative context schemes: historical self-enrollment and predictive TS-VAP use information through $k-1$, while synchronous ASD uses chunk k . TS-VAP anticipates target activity in k and beyond. (b) Color-matched examples of target-partner dialogue with unrelated interference; predicted activity is schematic.

pose an LLM-based TS-VAP that transfers a speech-LLM’s linguistic and conversational knowledge into predictive guidance, forecasting future activity directly from overlapping mixtures. (3) We systematically compare historical, synchronous, and predictive context for online AV-TSE, showing where LLM-derived prediction helps across multi-speaker conversation patterns.

2. METHOD

2.1. Problem Formulation

Let s_{tar} , s_{par} , and s_{ext} denote the target, conversational partner, and independent third-party speech, respectively. The single-channel mixture is

$$x[n] = s_{\text{tar}}[n] + s_{\text{par}}[n] + s_{\text{ext}}[n], \quad (1)$$

where n denotes the audio sample index, and the target and partner belong to the same dyadic dialogue.

For chunk-wise online processing, let x_k and v_k denote the mixture audio and target face track at step k . Target speaker extraction with predictive context is formulated as

$$\hat{s}_k = f_{\text{AV-TSE}}(x_k, v_k, C_{k-1}^P), \quad (2)$$

where $\hat{s}_k$ denotes the reconstructed target waveform at step k , and the predictive context C_{k-1}^P summarizes observations up to step $k-1$ to predict subsequent speaker activity.

2.2. LLM-based TS-VAP

We adapt Mini-Omni2 with a Qwen2-0.5B backbone into our target-speaker-conditioned voice activity projection module, TS-VAP. Whisper and CLIP encode the mixture audio x and target video v , respectively. Qwen produces visually conditioned audio representations, which are fused with the CLIP features before a lightweight causal head:

$$U = f_{\text{TS-VAP}}(x, v), \quad (3)$$

where U is a sequence of 256-dimensional predictive representations at 25 Hz, and U_ℓ denotes its ℓ -th frame. At each frame, $P_\ell = \text{Softmax}(WU_\ell + b)$ predicts joint target-partner activity over the next 2 s, encoded as $2^8 = 256$ joint states from four binary future-activity bins per speaker [10]. Training combines joint-state classification with auxiliary current-activity and future-bin supervision. After fine-tuning, TS-VAP is frozen, and U is used to construct the separator’s predictive context.

2.3. Online AV-TSE with Historical, Synchronous, and Predictive Knowledge

We evaluate three forms of context for conversational AV-TSE. **Historical context**, inspired by MeMo [17], summarizes past separator latents with a GRU to obtain C_{k-1}^H , which conditions the AV-fused features through a residual adapter. **Synchronous context**, inspired by ActiveExtract [8], combines ASD embeddings with features from its visual branch to form C_k^S . This cue is concatenated with face features and projected before AV fusion. **Predictive context** encodes anticipated target-partner activity from past TS-VAP features, as detailed below.

To construct the predictive context for chunk k , we use TS-VAP features U from the last 2 s before the current chunk, or all available history if shorter. We average the features within each 0.2-second interval to obtain at most ten tokens:

$$C_{k-1}^P = \text{Proj}(\text{LN}(\text{Pool}(U))) + E_{\text{rel}}, \quad (4)$$

where Pool is interval-wise mean pooling, LN is layer normalization, and Proj maps tokens to the audio feature dimension. E_{rel} provides learnable positional embeddings to distinguish recent from older history tokens.

Current audio bottleneck features A_k query this context:

$$R_k = \text{MHA}(\text{LN}(A_k), C_{k-1}^P, C_{k-1}^P), \quad (5)$$

$$\tilde{A}_k = \text{Conv}_{1 \times 1}([A_k; \text{LN}(R_k)]), \quad (6)$$

Table 1. Online extraction results under different backbones and contextual cues. Context is categorized as historical (History), synchronous (Sync.), and predictive (Pred.) information, corresponding to self-enrollment, ASD, and LLM-based TS-VAP prediction, respectively. RI denotes the relative improvement over the corresponding context-free baseline, higher is better.

Model	Context			TP SI-SNR by overlap ratio (dB) ↑						Avg. ↑	RI (%) ↑
	History	Sync.	Pred.	0%	(0, 20]%	(20, 40]%	(40, 60]%	(60, 80]%	(80, 100]%		
Baselines											
Dolphin [20] [†]	–	–	–	2.62	2.90	2.27	3.19	3.66	4.40	3.17	–
AV-TFGridNet [21] [†]	–	–	–	9.96	7.20	5.24	4.98	5.04	5.95	6.40	–
TDSE [22]	–	–	–	11.59	10.48	6.46	5.53	5.06	5.51	7.44	ref.
USEV [6]	–	–	–	14.07	11.59	8.13	7.03	6.18	7.24	9.04	ref.
AV-SepFormer [7]	–	–	–	15.56	11.27	7.91	6.71	6.39	6.58	9.07	ref.
LLM-VAP across Different Backbones											
TDSE [22]	–	–	✓	13.24	11.17	6.45	5.42	4.88	5.54	7.78	+4.6
AV-SepFormer [7]	–	–	✓	17.25	12.06	8.29	6.70	6.48	6.73	9.59	+5.7
USEV [6]	–	–	✓	15.67	12.26	8.38	7.23	6.60	7.20	9.56	+5.8
	–	–	✓ ^A	15.24	11.98	8.25	7.19	6.47	7.35	9.41	+4.1
Context Configurations on USEV											
USEV [6]	✓	–	–	16.08	12.23	8.33	7.15	6.59	7.05	9.57	+5.9
	–	✓	–	16.28	12.78	8.52	7.29	6.69	6.89	9.74	+7.8
	–	✓	✓	17.09	12.75	8.17	7.02	6.41	7.17	9.77	+8.1
	✓	✓	–	16.07	12.91	8.67	7.30	6.83	7.34	9.86	+9.1
	✓	–	✓	17.07	12.45	8.42	7.39	6.55	7.38	9.88	+9.3
	✓	✓	✓	17.19	13.25	8.52	7.27	6.84	7.09	10.03	+10.9

[†] Reference backbone without context adaptation. In Pred., ✓ denotes LLM-VAP and ✓^A denotes Acoustic VAP.

where MHA is four-head attention with arguments ordered as query, key, and value, and R_k is the retrieved context. Channel concatenation $[\cdot; \cdot]$ followed by a 1×1 convolution restores the audio feature dimension. The conditioned features $\tilde{A}_k$ enter AV fusion; when no valid context exists, $\tilde{A}_k = A_k$.

Two-pass training [17]. In the first pass, the separator processes the first τ seconds without predictive memory, and TS-VAP features from this prefix are pooled to construct the memory. In the second pass, the separator reprocesses the same prefix together with up to an additional 2 s, using the memory constructed from the first τ seconds and keeping it fixed.

2.4. Training Objective and Evaluation

Prior work handles target-absent mixtures by dividing them into discrete activity scenarios with hand-tuned loss weights [6, 8, 23]. We found this strategy less effective in our dialogue setting and instead add a personal voice activity detection (pVAD) head [24, 25], which predicts the frame-level probability p_t that the target speaker is active, providing explicit supervision over target-absent regions. The objective is

$$\mathcal{L} = \sum_{j=1}^2 \left(\mathcal{L}_{\text{ext}}^{(j)} + \mathcal{L}_{\text{pVAD}}^{(j)} \right), \quad (7)$$

where j indexes the two training passes, $\mathcal{L}_{\text{ext}}$ is the negative SI-SNR over target-active samples, scaled by their temporal proportion, and $\mathcal{L}_{\text{pVAD}}$ is binary cross-entropy. During inference, the target-activity probability p_t is linearly upsampled to waveform resolution and thresholded to construct an attenuation mask. The reconstructed target waveform is multiplied by this mask, with unit gain in predicted target-active regions and a fixed attenuation factor in predicted target-inactive regions.

We report SI-SNR for target-present parts, grouped by $r = |\mathcal{A}_{\text{tar}} \cap \mathcal{A}_{\text{int}}| / |\mathcal{A}_{\text{tar}} \cup \mathcal{A}_{\text{int}}|$, where $\mathcal{A}_{\text{int}}$ covers partner and third-

party activity.

3. EXPERIMENTAL SETUP

3.1. Datasets

Vox2Mix pretraining. We pretrain all AV-TSE backbones on two-speaker mixtures constructed from VoxCeleb2 [26]. Utterances from different speakers are onset-aligned, truncated to the shorter source, and mixed at a relative level sampled from $\mathcal{U}(-10, 10)$ dB, with mixture duration limited to 6 s. The training and validation sets contain 20,000 and 5,000 mixtures, respectively, from 800 speakers.

IEMOCAP-Dialog3Mix. We construct 6-s mixtures from IEMOCAP [18] by adding independent third-speaker interference to intact dyadic dialogues, preserving natural pauses, overlaps, and turn transitions. The interference duration is sampled from $\mathcal{U}(0, 6)$ s and its level from $\mathcal{U}(-5, 0)$ dB relative to the dialogue’s pooled active-speech RMS. Dialogue participants are disjoint across splits. The training, validation, and test sets contain 23,522, 8,120, and 3,000 mixtures, respectively.

RealTalk-Dialog3Mix. To evaluate cross-corpus generalization, we construct an additional 6-s test set from the unscripted, in-the-wild RealTalk conversations [19]. We retain annotated dyadic speech in its original timing and add third-speaker interference following the same procedure as above. The set contains 1,000 mixtures from 500 dialogue windows, with both participants used as targets.

3.2. Implementation Details

We compare five representative AV-TSE extractors spanning different architectures: Dolphin [20], a lightweight audio-visual separation model; AV-TFGridNet [21], a time-frequency GridNet-based extractor; AV-SepFormer [7], a transformer-based extractor; TDSE [22], a convolution-based extractor; and USEV [6], a widely used dual-path RNN baseline. All are adapted to the same causal

Table 2. Correct suppression rate (CSR, %) under non-target interference when the target is silent. Higher is better.

Model	Partner interference	Third-party interference
USEV baseline	83.96	79.73
USEV + Acoustic VAP	85.71	81.86
USEV + LLM-VAP	87.74	85.13

online setting, initialized from Vox2Mix pretraining, and trained under the same protocol. We evaluate TS-VAP on AV-SepFormer, TDSE, and USEV to test its consistency across architectures, and run all context combinations on USEV. Every system uses a pVAD head.

Following [12], TS-VAP is trained separately using 10-s observations from the IEMOCAP training split to predict the subsequent 2 s. Its task-specific layers are first trained with Mini-Omni2 frozen at 3×10^{-4} for up to six epochs; we then jointly optimize these layers and rank-8 Qwen LoRA adapters at 10^{-4} and 3×10^{-5} , respectively, for up to eight epochs. TS-VAP remains frozen during AV-TSE training, where all separators use Adam at 10^{-4} , a batch size of 32, and equal weights for both passes, trained for up to 100 epochs with a patience of ten epochs. Online inference uses a 2-s initialization/processing window and a 600-ms hop, with the pVAD threshold and inactive-region attenuation set to 0.4 and 0.05, respectively, based on validation performance. For comparison, Acoustic VAP uses a TalkNet-based audio-only architecture inspired by [12] with the same training protocol as TS-VAP.

4. RESULTS

4.1. Core Results

Table 1 summarizes the online extraction results on 3,000 IEMOCAP-Dialog3Mix test mixtures. All systems use the same pVAD module and a 2-s processing window with a 600-ms hop for online inference. Across the three backbones, TS-VAP consistently improves the average TP SI-SNR, with relative gains of 5.7%, 4.6%, and 5.8% for AV-SepFormer, TDSE, and USEV, respectively. On USEV, TS-VAP achieves 9.56 dB, outperforming Acoustic VAP by 0.15 dB, indicating that target-conditioned audio-visual prediction provides additional information beyond acoustic prediction alone. USEV shows the largest relative gain from TS-VAP and is therefore adopted for the subsequent context-configuration experiments.

4.2. Effectiveness of Historical, Synchronous, and Predictive Context

We further evaluate historical, synchronous, and predictive context on USEV. Among the individual cues, synchronous context performs best, while all three consistently improve over the context-free baseline. Among pairwise combinations, History+Pred. achieves the best performance, slightly outperforming History+Sync. and Sync.+Pred., suggesting that predictive context is more complementary to historical speaker information than to synchronous activity cues. Combining all three contexts yields the best overall average of 10.03 dB, corresponding to a 10.9% relative improvement over the context-free baseline. The full configuration, however, provides no gain in the (80, 100] % overlap bin, suggesting that contextual cues are most useful when turn-taking structure remains exploitable.

4.3. Interference Suppression Guided by TS-VAP

Target-silent regions are challenging for TSE because active non-target speech can easily leak into the output. This is particularly dif-

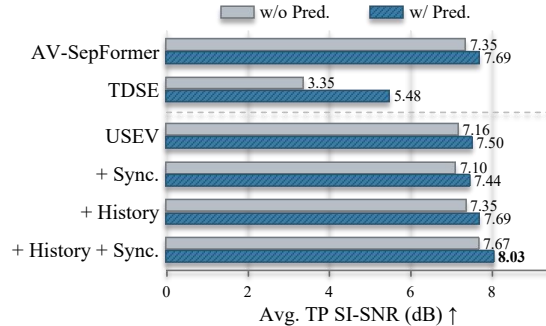


Fig. 3. Zero-shot online extraction on the 1,000-mixture RealTalk subset. Each configuration is evaluated with and without the LLM-based TS-VAP. TP SI-SNR in dB; AV-SepFormer and TDSE report only the baseline, while USEV also includes sync., history, and their combination.

ficult for unrelated third-party interference. Predictive context from TS-VAP may help the separator suppress such leakage by providing cues about the expected activity of the ongoing conversation.

We evaluate this behavior on all 3,000 IEMOCAP test mixtures under *partner interference* and *third-party interference*, where the corresponding non-target speaker is active while the target is silent. The two conditions may overlap when both interferers are active. We report the *correct suppression rate* (CSR), measuring how often the extractor correctly suppresses its output in these target-silent regions.

As shown in Table 2, Acoustic VAP improves CSR over the baseline under both interference conditions, while LLM-VAP provides further gains of 2.03 and 3.27 percentage points for partner and third-party interference, respectively. Compared with the baseline, LLM-VAP improves CSR by 3.78 and 5.40 percentage points. The larger gain under third-party interference suggests that target-conditioned audio-visual predictive context can improve robustness to unrelated speech outside the target-partner conversational structure.

4.4. Zero-shot Transfer to RealTalk

We further evaluate zero-shot transfer to the RealTalk subset under the same online protocol. As shown in Fig. 3, TS-VAP improves all tested backbones and context configurations. The gain is particularly pronounced for TDSE, increasing from 3.35 to 5.48 dB, while AV-SepFormer and USEV improve from 7.35 to 7.69 dB and from 7.16 to 7.50 dB, respectively. Predictive context also remains beneficial when combined with historical and synchronous context, with the full-context configuration achieving the best result of 8.03 dB. These results demonstrate that the benefits of TS-VAP persist under the RealTalk domain shift.

5. CONCLUSION

We studied streaming AV-TSE in natural conversations with third-party interference and proposed an LLM-based TS-VAP to provide predictive conversational context. Results show that predictive context improves extraction and complements historical and synchronous cues; moreover, the LLM-based TS-VAP outperforms its acoustic counterpart, and jointly leveraging all three context types yields the best performance. Future work will integrate the LLM more deeply into the extraction pipeline, moving from predictive conditioning toward fully semantic-driven separation.

6. REFERENCES

- [1] OpenAI, “GPT-4o system card,” *arXiv preprint arXiv:2410.21276*, 2024.
- [2] Katerina Zmolikova, Marc Delcroix, Tsubasa Ochiai, Keisuke Kinoshita, Jan Cernocky, and Dong Yu, “Neural target speech extraction: An overview,” *IEEE Signal Processing Magazine*, vol. 40, no. 3, pp. 8–29, 2023.
- [3] Wenxuan Wu, Xueyuan Chen, Shuai Wang, Jiadong Wang, Lingwei Meng, Xixin Wu, Helen Meng, and Haizhou Li, “ C^2 AV-TSE: Context and confidence-aware audio visual target speaker extraction,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 19, no. 4, pp. 646–657, 2025.
- [4] Wenxuan Wu, Xueyuan Chen, Xixin Wu, Haizhou Li, and Helen Meng, “Target speech extraction with pre-trained AV-HuBERT and mask-and-recover strategy,” in *International Joint Conference on Neural Networks (IJCNN)*, 2024, pp. 1–8.
- [5] Ariel Ephrat, Inbar Mosseri, Oran Lang, Tali Dekel, Kevin Wilson, Avinatan Hassidim, William T. Freeman, and Michael Rubinstein, “Looking to listen at the cocktail party: A speaker-independent audio-visual model for speech separation,” *ACM Transactions on Graphics*, vol. 37, no. 4, pp. 112:1–112:11, 2018.
- [6] Zexu Pan, Meng Ge, and Haizhou Li, “USEV: Universal speaker extraction with visual cue,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 3032–3045, 2022.
- [7] Jiuxin Lin, Xinyu Cai, Heinrich Dinkel, Jun Chen, Zhiyong Yan, Yongqing Wang, Junbo Zhang, Zhiyong Wu, Yujun Wang, and Helen Meng, “AV-SepFormer: Cross-attention SepFormer for audio-visual target speaker extraction,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5.
- [8] Junjie Li, Ruijie Tao, Zexu Pan, Meng Ge, Shuai Wang, and Haizhou Li, “Audio-visual active speaker extraction for sparsely overlapped multi-talker speech,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024, pp. 10666–10670.
- [9] Zexu Pan, Wupeng Wang, Shengkui Zhao, Chong Zhang, Kun Zhou, Yukun Ma, and Bin Ma, “Online audio-visual autoregressive speaker extraction,” in *Proc. Interspeech*, 2025, pp. 1928–1932.
- [10] Erik Ekstedt and Gabriel Skantze, “Voice activity projection: Self-supervised learning of turn-taking events,” in *Proc. Interspeech*, 2022, pp. 5190–5194.
- [11] Koji Inoue, Bing’er Jiang, Erik Ekstedt, Tatsuya Kawahara, and Gabriel Skantze, “Real-time and continuous turn-taking prediction using voice activity projection,” in *International Workshop on Spoken Dialogue Systems Technology (IWSDS)*, 2024, Demo paper.
- [12] Antonio Cano, Guillermo Pérez, Luis Merino, and Randy Gomez, “Multimodal voice activity projection for turn-taking in social robots with voice-activity-related pretrained encoders,” *arXiv preprint arXiv:2607.07294*, 2026.
- [13] Hyunbae Jeon, Frederic Guintu, and Rayvant Sahni, “Lla-VAP: LSTM ensemble of Llama and VAP for turn-taking prediction,” *arXiv preprint arXiv:2412.18061*, 2024.
- [14] Wenxuan Wu, Shuai Wang, Xixin Wu, Helen Meng, and Haizhou Li, “Incorporating linguistic constraints from external knowledge source for audio-visual target speech extraction,” in *Proc. Interspeech*, 2025, pp. 5053–5057.
- [15] Wenxuan Wu, Shuai Wang, Xixin Wu, Helen Meng, and Haizhou Li, “ELEGANCE: Efficient LLM guidance for audio-visual target speech extraction,” *IEEE Transactions on Audio, Speech and Language Processing*, vol. 34, pp. 2801–2814, 2026.
- [16] Zhifei Xie and Changqiao Wu, “Mini-Omni2: Towards open-source GPT-4o with vision, speech and duplex capabilities,” *arXiv preprint arXiv:2410.11190*, 2024.
- [17] Junjie Li, Wenxuan Wu, Shuai Wang, Zexu Pan, Kong Aik Lee, Helen Meng, and Haizhou Li, “MeMo: Attentional momentum for real-time audio-visual target speaker extraction under impaired visual conditions,” *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 34, pp. 3491–3504, 2026.
- [18] Carlos Busso, Murtaza Bulut, Chi-Chun Lee, Abe Kazemzadeh, Emily Mower, Samuel Kim, Jeannette N. Chang, Sungbok Lee, and Shrikanth S. Narayanan, “IEMO-CAP: Interactive emotional dyadic motion capture database,” *Language Resources and Evaluation*, vol. 42, no. 4, pp. 335–359, 2008.
- [19] Scott Geng, Revant Teotia, Purva Tendulkar, Sachit Menon, and Carl Vondrick, “Affective faces for goal-driven dyadic communication,” *arXiv preprint arXiv:2301.10939*, 2023.
- [20] Kai Li, Kejun Gao, and Xiaolin Hu, “Efficient audio-visual speech separation with discrete lip semantics and multi-scale global-local attention,” in *International Conference on Learning Representations (ICLR)*, 2026.
- [21] Zexu Pan, Gordon Wichern, Yoshiki Masuyama, François G. Germain, Sameer Khurana, Chiori Hori, and Jonathan Le Roux, “Scenario-aware audio-visual TF-GridNet for target speech extraction,” in *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2023, pp. 1–8.
- [22] Jian Wu, Yong Xu, Shi-Xiong Zhang, Lianwu Chen, Meng Yu, Lei Xie, and Dong Yu, “Time domain audio visual speech separation,” in *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2019, pp. 667–673.
- [23] Bang Zeng and Ming Li, “Universal speaker embedding free target speaker extraction and personal voice activity detection,” *Computer Speech & Language*, vol. 94, pp. 101807, 2025.
- [24] Qingjian Lin, Lin Yang, Xuyang Wang, Luyuan Xie, Chen Jia, and Junjie Wang, “Sparsely overlapped speech training in the time domain: Joint learning of target speech separation and personal vad benefits,” in *Proc. APSIPA ASC*, 2021, pp. 689–693.
- [25] Shaojin Ding, Quan Wang, Shuo-Yiin Chang, Li Wan, and Ignacio Lopez Moreno, “Personal vad: Speaker-conditioned voice activity detection,” in *Proc. Odyssey*, 2020, pp. 433–439.
- [26] Joon Son Chung, Arsha Nagrani, and Andrew Senior, “VoxCeleb2: Deep speaker recognition,” in *Proc. Interspeech*, 2018, pp. 1086–1090.